Biochemistry Cume

April 4, 2014

1. 60 points. For each peptide or amino acid below, estimate the net charge (to ± 0.5 atomic charge) at pH 10 and at pH 4, and estimate the isoelectric point (to ± 0.5 pH unit).

	Molecule	charge at pH 10	charge at pH 4	approx. pI
a.	Gly	___	___	___
b.	Arg	___	___	___
c.	Gly-Arg	___	___	___
d.	Gly-Arg-Arg	___	___	___
e.	Arg-Gly-Arg	___	___	___
f.	Arg-Gly-Arg-Arg	___	___	___
g.	Asp	___	___	___
h.	Gly-Asp	___	___	___
i.	Gly-Asp-Arg	___	___	___
j.	Gly-Asp-Asp-Arg	___	___	___

2. 30 points. What single method would you suggest to achieve the best possible separation of as many as possible of the molecules in question 1 from a mixture of all of them? (Assume that you are working on a 10-milligram scale.) Describe your method in as much detail as time and knowledge permit. What equipment, materials and supplies will you need? How long will it take to do the separation? What is the chemical or physical basis for the separation? What advantages are offered by your method over other possible choices?

3. 10 points. [More than one answer may be possible.] Which pair of molecules in question one should be the most difficult to separate from each other? Why? Supposing that they were separated, what analytical method could be used to distinguish these two molecules?

Abbreviations: Asp, aspartic acid; Arg, arginine; Gly, glycine.